实验报告

实验一

```
Last login: Tue Apr 21 14:46:42 on ttys013
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook code % cd .. # 去到某个路径
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % pwd # 显示当前路径
/Users/yzhbradoodrrpurp/Desktop/OS
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % ls # -a 显示所有文件; -l 显示具体
信息
code  notes slides
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % touch test.txt
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % cp test.txt backup.txt # 复制文
件; -r 递归复制
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % mv test.txt ./code/text.txt # -
r 递归删除
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % rm backup.txt # -r 递归删除
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % mkdir test
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % rmdir test
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % cat code/text.txt
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % more code/lab-gcc-make/myapp.c
# 分页查看
#include "greeting.h"

int main(void) {
    greeting();
    return 0;
}
 ESCOC









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 ESCOD
#include "greeting.h"

int main(void) {
    greeting();
    return 0;
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 ESCOD
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    greeting();
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 ESCOD
#include "greeting.h"

int main(void) {
    greeting();
    return 0;
}
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yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % less code/lab-gcc-make/myapp.c
# 分页查看
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % head code/lab-gcc-make/myapp.c
# tail file 查看前10行; -n 20 查看前20行
#include "greeting.h"

int main(void) {
    greeting();
    return 0;
}
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % tail code/lab-gcc-make/myapp.c
# 查看后10行; -n 20 查看后20行
```

```
#include "greeting.h"

int main(void) {
    greeting();
    return 0;
}
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % find . -name "*.txt" # 根据名字找文件
./code/text.txt
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % whereis gcc # 查找命令
gcc: /usr/bin/gcc
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % grep "main" ./code/lab-gcc-make/myapp.c # 在文件中搜索字符串
int main(void) {
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % chmod u+w ./code/lab-gcc-make/myapp.c # 对于该文件给用户加上写权限

yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % tar -czvf ./code/lab-gcc-make.tar.gz ./code/lab-gcc-make # c=创建 z=gzip v=显示过程 f=文件名
a ./code/lab-gcc-make
a ./code/lab-gcc-make/hello.c
a ./code/lab-gcc-make/experiment.md
a ./code/lab-gcc-make/.DS_Store
a ./code/lab-gcc-make/Makefile
a ./code/lab-gcc-make/resources
a ./code/lab-gcc-make/myapp.c
a ./code/lab-gcc-make/greeting.c
a ./code/lab-gcc-make/myapp
a ./code/lab-gcc-make/hello.i
a ./code/lab-gcc-make/hello.s
a ./code/lab-gcc-make/hello.o
a ./code/lab-gcc-make/hello
a ./code/lab-gcc-make/experiment.pdf
a ./code/lab-gcc-make/greeting.o
a ./code/lab-gcc-make/greeting.h
a ./code/lab-gcc-make/myapp.o
a ./code/lab-gcc-make/resources/make.png
a ./code/lab-gcc-make/resources/hello.png
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % gzip ./code/text.txt
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % touch test.txt
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS % bzip2 test.txt
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook OS %
```

Hello gcc

hello.c 文件内容如下：

```c
#include <stdio.h>

int main(void) {
    printf("Hello gcc!\n");
    return 0;
}
```

在 Terminal 中通过如下指令得到 *.c, *.i, *.s, *.o 文件：

```bash
# 预处理：.c -> .i
gcc -E hello.c -o hello.i
# 编译成汇编：.i/.c -> .s
gcc -S hello.c -o hello.s
# 汇编成目标文件：.s/.c -> .o
gcc -c hello.c -o hello.o
# 链接成可执行文件
gcc hello.c -o hello
```

Terminal 截图：

```
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook lab-gcc-make % gcc -E hello.c -o hello.i
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook lab-gcc-make % gcc -S hello.c -o hello.s
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook lab-gcc-make % gcc -c hello.c -o hello.o
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook lab-gcc-make % gcc hello.c -o hello
```

Makefile

myapp.c 文件内容：

```c
#include "greeting.h"

int main(void) {
    greeting();
    return 0;
}
```

greeting.c 文件内容：

```c
#include <stdio.h>
```

```c
#include "greeting.h"

void greeting(void) {
    printf("Hello from greeting.c!\n");
}
```

greeting.h 文件内容:

```c
#ifndef GREETING_H
#define GREETING_H

void greeting(void);

#endif
```

Makefile 文件内容:

```makefile
CC = gcc
CFLAGS = -Wall -Wextra -g


.PHONY: all run hello_stages clean

all: myapp

myapp: myapp.o greeting.o
  $(CC) -o $@ $^

myapp.o: myapp.c greeting.h
  $(CC) $(CFLAGS) -c myapp.c -o myapp.o

greeting.o: greeting.c greeting.h
  $(CC) $(CFLAGS) -c greeting.c -o greeting.o

run: myapp
  ./myapp

hello_stages: hello.c
  $(CC) -E hello.c -o hello.i
  $(CC) -S hello.c -o hello.s
  $(CC) -c hello.c -o hello.o
  $(CC) hello.c -o hello
  $(CC) -g hello.c -o hello_g

clean:
  rm -f *.o myapp hello hello_g hello.i hello.s
```

在 Terminal 当前目录下输入 make 即可得到可执行文件:

```
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook lab-gcc-make % make
gcc -Wall -Wextra -g -c myapp.c -o myapp.o
gcc -Wall -Wextra -g -c greeting.c -o greeting.o
gcc -o myapp myapp.o greeting.o
yzhbradoodrrpurp@yzhbradoodrrpurps-MacBook lab-gcc-make %
```